



Programming Fundamentals

LAB # 1

sec: b8

Instructions:

- You have to submit CPP files of the program in a folder, named your Registration Number.
- You have to upload your lab tasks on CMS.
- All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- Indent and comment your code.
- Use meaningful variable names
- Plan your code carefully on a piece of paper before you implement it.

Learning Objectives:

- Demonstrate knowledge of basic **file handling** programming C++.
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1. Write a C++ program that opens a file named "data.txt", reads only the first line (header), and displays it.

The program should:

- Open the file "data.txt" for reading.
- Check if the file exists and handle errors if it cannot be opened.
- Read only the first line from the file (header).
- Display the header on the screen.
- Close the file after reading

Expected Output:

Sample Input File (data.txt):

Student Name, Age, Grade

Ali, 20, A

Sara, 19, B

Expected Output:

Header: Student Name, Age, Grade

2: Write a C++ program that performs the following operations:

- Open a file named "input.txt" for reading.

- *Check if the file exists and handle errors if it cannot be opened.*
- *Read the file character by character.*
- *Display each character on the screen as it is read.*
- *Close the file after reading all characters.*

Sample Input File (input.txt):

Hello!

File content displayed on console:

Hello!

3. Write a C++ program that performs the following operations:

- *Open a file named "text.txt" for reading.*
- *Check if the file exists and handle errors if it cannot be opened.*
- *Read the file line by line and count the total number of lines.*
- *Display the total number of lines in the file.*
- *If the file is empty, display an appropriate message.*
- *Close the file after reading*

Expected Output:

Sample Input File (text.txt):

This is the first line.

This is the second line.

This is the third line.

Expected Output:

Total lines: 3

4. Write a C++ program that opens "source.txt" for reading and "destination.txt" for writing.

- The program should copy the contents of "source.txt" to "destination.txt" character by character.

Output:

Sample Input File (source.txt):

C++ file handling is easy!

Expected Output in destination.txt:

C++ file handling is easy!

Console Output:

File copied successfully!

5: Write a C++ program that:

- *Opens a file named "data.txt" for reading.*
- *Reads and displays only the first character from the file.*
- *If the file does not exist, print an error message.*
- *If the file is empty, print a message indicating that it is empty.*
- *Closes the file after reading.*

Sample Input File (data.txt):

Hello World!

Output:

First character: H

End of LAB 3 😊